

# Gadget For DUAL PROTECTION FROM VIRUSES

RAKESH JAIN

The Corona pandemic is over officially but its ill effects are still being seen, especially in crowded places like popular markets and malls. Besides, one needs to guard against new viruses and mutations of Corona.

The World Health Organization (WHO) had recommended keeping two-metre distance from others and rubbing hands with soap for at least 20 seconds before washing, frequently, as a safeguard. The proposed de-

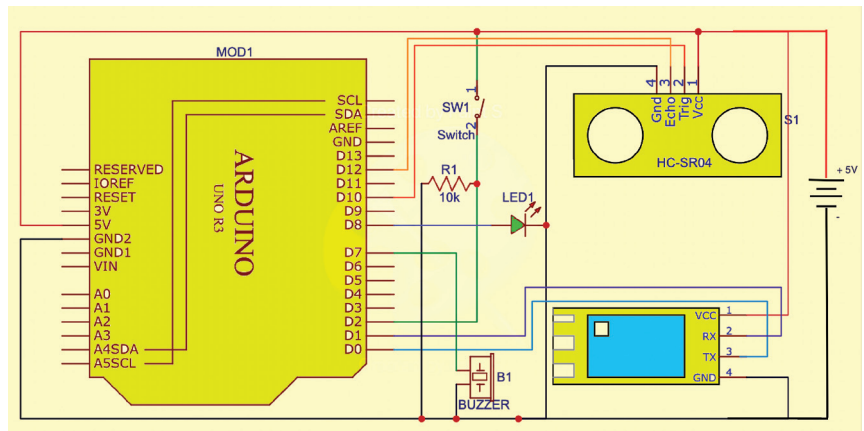


Fig. 1: Circuit diagram

**TABLE 1**  
**BILL OF MATERIALS**

| Name                           | Quantity |
|--------------------------------|----------|
| Arduino Uno (MOD1)             | 1        |
| Ultrasonic sensor HC-SR04 (S1) | 1        |
| HC-05 Bluetooth (MOD2)         | 1        |
| Buzzer (B1)                    | 1        |
| LED1                           | 1        |
| Push-to-on switch (SW1)        | 1        |
| 5V battery/5V adaptor          | 1        |
| Battery holder                 | 1        |

**TABLE 2**  
**CONNECTIONS BETWEEN  
THE ARDUINO UNO BOARD  
AND THE DEVICE**

| Arduino Uno Pin | Device Connected                                            |
|-----------------|-------------------------------------------------------------|
| D12             | Echo pin of ultrasonic sensor                               |
| D10             | Trig pin of ultrasonic sensor                               |
| D8              | Anode pin of green LED                                      |
| D13             | Anode pin red LED internally connected in Arduino Uno board |
| D7              | Positive of buzzer                                          |
| D2              | Push button switch                                          |
| Tx-D1           | RXD pin of HC-05 Bluetooth module                           |
| Rx-D0           | TXD pin of HC-05 Bluetooth module                           |

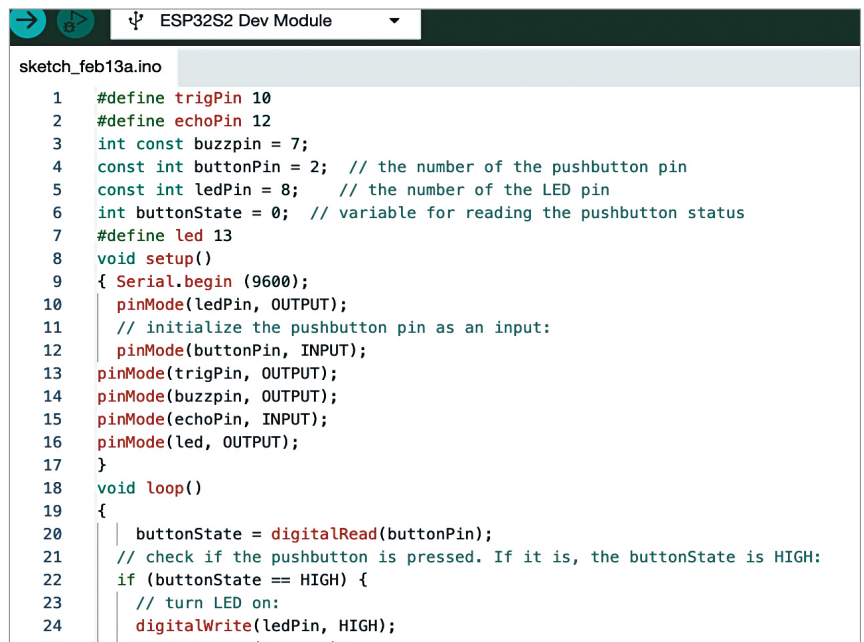


Fig. 2: Screenshot of the source code

vice can help in following both the recommendations of WHO.

Components required to build this gadget are listed in the Bill of Materials Table 1.

## Circuit and working

The circuit diagram of the gadget is shown in Fig. 1, while Table 2 shows the connections between the